ADHITYA THIRUMALA

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Education

University of Illinois Urbana-Champaign - Siebel School of Computer Science **GPA: 3.86**
Major: Computer Science + Linguistics, Minor: Mathematics Aug 2024 – May 2028

Experience

Cloudflare **May 2026 – Aug 2026**
Software Engineering Intern *Austin, TX*

- Developed test-generation tool in Rust for distributed synthetic end-to-end probe platform (Flamingo) on edge network that runs **750M+** tests per day to support health-mediated software releases, Service Level Objective (SLO) monitoring, and company-wide resiliency efforts
- Implemented internal/external-facing API fuzzing service in Rust that utilized edge-network for massively parallel test running over **60k+** nodes over **1375** data centers
- Used tool to determine health of Kubernetes service region-failover status to support high availability (HA) API deployment
- Extended Kubernetes-based test-scheduling control plane in Go to support bespoke test-scheduling needs for fuzzing service

University of Illinois Siebel School of Computer Science **Jan 2026 – Present**
Course Assistant *Urbana, IL*

- Assisted students in understanding and learning data structures in C++ (graphs, trees, linked lists) through office hours and online forum
- Assisted students in learning systems programming concepts in C (synchronization, concurrency, networking, filesystems, scheduling)
- Developed exam programming questions, test cases, solutions on PrairieLearn platform

Digital Main **May 2025 – Aug 2025**
Machine Learning Intern *Sunnyvale, CA*

- Built an asynchronous agentic retrieval-augmented-generation (RAG) system in Python, orchestrating query decomposition and research across web and internal client data (Slack messages, emails, etc.) using LangGraph
- Utilized FastAPI to create customer-facing production API to stream research process to web client using Server-sent Events (SSE)

Research and Projects

Distributed Filesystem - Rust, Tokio, Tonic, gRPC, Protobuf

- Built distributed filesystem in Rust based on Hadoop Distributed Filesystem (HDFS) using Tokio and gRPC (Tonic) that scales to 50+ nodes
- Implemented distributed consensus protocol (Raft) and key-value store to provide foundations for filesystem
- Designed gossip-style health-checking system with cancellable timers for failure detection
- Developed distributed grep tool to filter for logs emitted by above systems

C Compiler (Github)

- Implemented C compiler (C17 Standard) in Rust with hand-rolled recursive-descent parser, iterator-based (lazy) lexer, semantic analysis pass, and x86 code generation for macOS and Linux
- Verified compiler against comprehensive test case suite of **400+** programs to test functionality

Clickbait Classification and Spoiling Using Natural Language Processing (arXiv)

- Performed analysis between fine-tuned SLM BERT-based models and LLMs on clickbait spoiling shared task to achieve **80+%** accuracy
- Results accepted by *The 17th International Workshop on Semantic Evaluation*: Toronto, Canada 2023

Extractive Question Answering(QA) on Queries in Hindi and Tamil (arXiv)

- Trained NLP models in Google competition to perform extractive QA increasing accuracy by **20%** over baseline
- Received 26th Annual Carnegie Science Award given to **12** top scientists, entrepreneurs, and innovators across the city of Pittsburgh

Leadership

Association for Computing Machinery at Illinois **Mar 2025 – Mar 2026**
Treasurer *Urbana, IL*

- Managed over **$350k** of organizational funds for largest CS student organization at Illinois with **1000+** total members
- Migrated accounting and credit card platforms from BILL.com to Ramp and Quickbooks to obtain credit limit increase of **2800%** and to streamline reimbursements, onboarding, and reporting workflows

Awards and Competitions

UIUC ACM SIGPwny FallCTF: 6th Place (Advanced Track) in cybersecurity competition across 100+ teams from UIUC

Skills

Languages: Rust, C++, C, Python, Bash, LaTeX, Typst, SQL, Go, Lua, JavaScript, Typescript, CSS, Verilog, MIPS Assembly, Haskell
Tools: Linux, Git, Vim, Huggingface, Docker, Docker Compose, Kubernetes, PostgreSQL, AWS, Oracle Cloud, Google Cloud Platform, Cloudflare (Networking, Developer Platform), Github Actions CI/CD, Ghidra, Cargo, Bun, Prometheus, Grafana, Terraform, Vault (Hashicorp), MLIR, JAX, LLVM, PyTorch, gRPC, Protobuf, Cap'n proto
Libraries: Axum, Tokio, tonic, pandas, NumPy, Matplotlib, Tensorflow, pyTorch, Sci-Kit Learn, React, Flask, LangChain, LangGraph, FastAPI

Relevant Coursework

Computer Science: Distributed Systems, Compiler Construction, Programming Languages and Compilers, ML Compilers (Graduate), System Programming, Computer Systems Organization, Data Management in the Cloud, Algorithms and Models of Computation, Data Structures in C++, Computer Architecture, Computational Linguistics
Mathematics and Statistics: Honors Abstract Linear Algebra, Statistics and Probability, Introduction to Higher Level Math, Multivariable Calculus, Discrete Structures